Password Management Requirement Analysis

Group 5

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Persona 1: Eli Rincon

Summary

- Age: 38
- Department: IT Consulting & Support

Background

- 38-year-old IT consultant working across multiple client environments
- Tech-savvy, manages personal and work credentials
- Resets passwords monthly
- Uses a password manager regularly with confidence

Goals

- Seamless blend of convenience and security
- Biometric login (fingerprint)
- Organized storage with categories
- Secure storage for 2FA keys
- Auto-generated strong passwords
- Cross-platform autofill
- Emergency access for trusted contacts
- Affordable pricing

Pain Points

- Endless reset loop when passwords appear “already in use”
- Conflicts between personal and employer managers using the same service
- Difficulty managing multiple vaults without conflict detection
- Balancing varying security requirements across clients

Tech Confidence : Very Confident

Persona 2: Paul Thompson

Summary

- Age: 52
- Department: Electrical Engineering

Background

- 52-year-old engineer
- Resets passwords only a few times per year
- Stores passwords in a basic document or printed form
- Somewhat confident with technology; prefers simple solutions
- Open to new tools with clear benefits

Goals

- Easy retrieval when needed
- Emergency access for family or trusted contacts
- Clear legacy/succession setup
- Simple, guided onboarding
- Minimize password reset frequency

Pain Points

- Frequently forgets passwords and must reset
- Current low-tech storage creates security risks
- Unsure how to plan for emergency or legacy access
- Needs clear tutorials and step-by-step guidance

Tech Confidence : Somewhat Confident

Scenario For Paul

Paul is preparing documentation for a project when he needs to log into a vendor portal he hasn't accessed in several months. He opens the document where he usually keeps his passwords, but the file is cluttered and several entries are outdated. After trying a few guesses, he realizes he will need to reset the password again, something that has become a recurring inconvenience.

With the new notepad-style password manager, Paul creates a new entry labeled Vendor Portal and adds the updated password along with the security question and answer. The interface is simple enough that he doesn't have to learn a new system; it works like a structured notes app. A few days later, when he needs the same login again, he uses the Search bar to type "vendor," and the entry appears immediately. Everything he needs, password, security question, and notes, is stored in one place. The tool reduces the time he spends searching through documents and helps him avoid repeated resets.

Scenario For Eli

Eli is updating credentials for one of his consulting clients. The system requires a strong password with specific complexity rules. Normally, he would open a separate password-generator website, copy the result, and paste it into a document or his existing password manager, hoping it doesn't conflict with his employer's tools.

Using the new password manager, Eli selects Generate Password, reviews the suggested option, and saves it directly into a new entry labeled Client Admin Portal. He adds the client's security questions and answers in the same entry so he doesn't have to track them separately. Later, when he needs to access the portal again, he types "admin" into the Search bar and retrieves the entry instantly. The notepad-style layout keeps everything straightforward, and the separation between entries prevents conflicts with his employer's systems. The tool supports his security standards without adding unnecessary complexity.

Functional Requirements

FR1 - Create and Store Entries

Requirement ID: FR1
Type: Functional
Description: The system shall allow users to create a new entry containing a title, username, password, and notes
Rationale: Paul currently stores passwords in an outdated document and needs a reliable place to save updated credentials.
Fit Criterion: Users must be able to create and save a new entry in under 30 seconds during usability.

FR2 - Search for Saved Entries

Requirement ID: FR2
Type: Functional
Description: The system shall provide a search function that returns matching entries based on keywords in the title or notes.
Rationale: Paul needs to quickly retrieve specific accounts from a large set of 40+ logins.
Fit Criterion: Search results must appear within 1 second and return the correct entry 95% of the time.

FR3 - Password Generator

Requirement ID: FR3
Type: Functional
Description: The system shall generate strong, random passwords that meet standard complexity requirements.
Rationale: Eli frequently creates new passwords for clients and needs a built-in generator to avoid using external tools.
Fit Criterion: Generated passwords must include uppercase, lowercase, numbers, and symbols, and be at least 12 characters long.

FR4 - Edit and Update Existing Entries

Requirement ID: FR4
Type: Functional
Description: The system shall allow users to edit any field within an existing entry, including password and security questions.

Rationale: Paul often resets passwords and needs a simple way to update entries without creating duplicates.
Fit Criterion: Users must be able to update an entry in under 15 seconds with no more than three steps.

FR5 - Entry Sorting

Requirement ID: FR5
Type: Functional
Description: The system shall allow users to sort entries alphabetically or by date created.
Rationale: Paul manages 40+ accounts and needs simple organization tools beyond search.
Fit Criterion: Sorting must update the list within 1 second and correctly reorder entries 100% of the time.

Non-Functional Requirements

NFR1 - Usability

Requirement ID: NFR1
Type: Non-Functional
Description: The system shall have a simple, notepad-style interface that allows first-time users to create and find entries without training.
Rationale: Paul prefers simple tools and becomes frustrated with complex systems.
Fit Criterion: At least 85% of first-time users must complete a “create and search for an entry” task within 2 minutes.

NFR2 - Reliability

Requirement ID: NFR2
Type: Non-Functional
Description: The system shall maintain consistent performance without data loss during normal use.
Rationale: Eli manages critical client credentials and cannot risk losing stored information.
Fit Criterion: No data loss incidents may occur during 30 consecutive days of simulated use.

NFR3 - Performance

Requirement ID: NFR3
Type: Non-Functional
Description: The system shall load the full list of entries and the search interface within 1 second.
Rationale: Paul and Eli both rely on quick access to credentials during work tasks.
Fit Criterion: Interface load time must not exceed 1 second on standard hardware.

Requirements Critique

Critique 1

Before: “The app should be simple to use”
After: “At least 85% of first-time users must complete a ‘create and search for an entry’ task within 2 minutes.”
Explanation: The original statement was subjective and vague. The revised version defines a measurable usability threshold and a specific task, making it testable.

Critique 2

Before: “The system should generate strong passwords”
After: “Generated passwords must include uppercase, lowercase, numbers, and symbols, and be at least 12 characters long.”
Explanation: The original requirement did not define what “strong” meant. The refined version specifies complexity rules and length, making the requirement objective and verifiable.